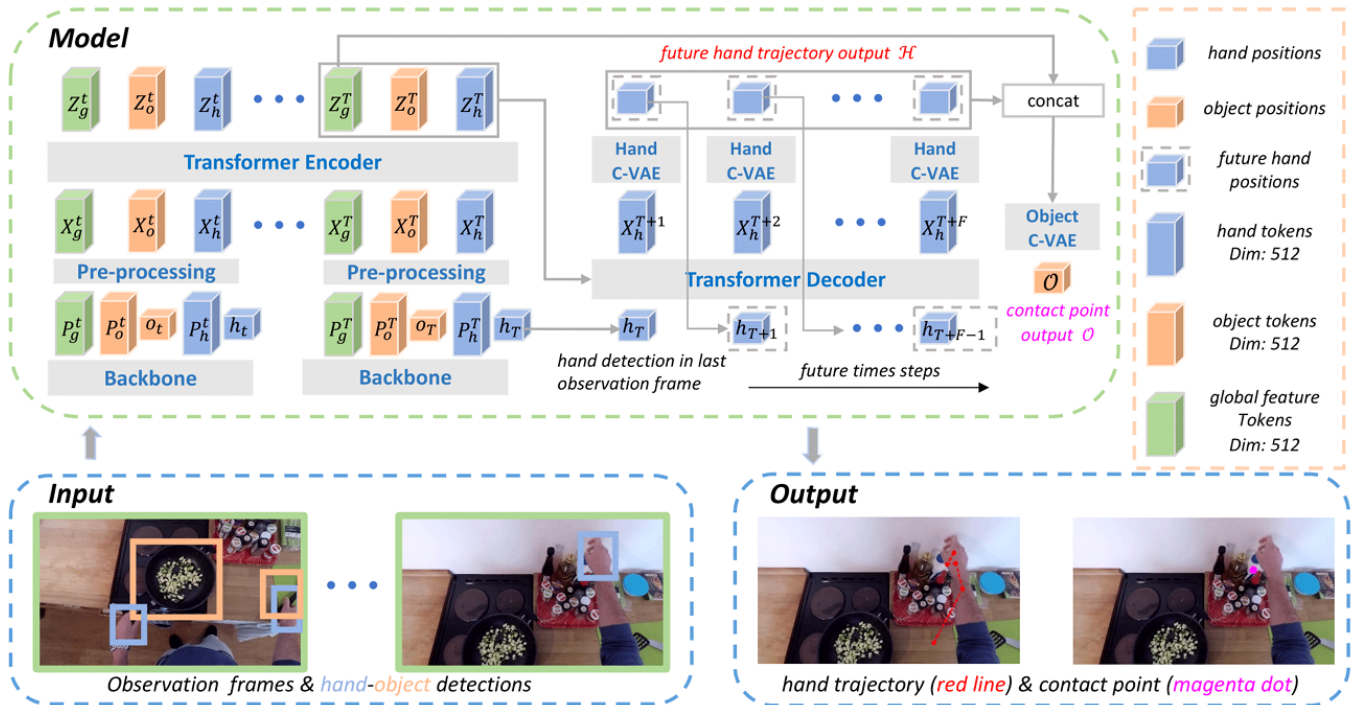


阶段总结

受论文[<https://arxiv.org/pdf/2204.01696>](Joint Hand Motion and Interaction Hotspots Prediction from Egocentric Videos)的启发



- 原版论文先从观察帧里提取 object-centric tokens，每帧会检测hand bbox，object bbox，global scene feature
- 然后用ConvNet + RoIAlign 提取特征：hand token，object token，global token。将整张图的场景拆成了手，物体，全局环境三个token
- 用Transformer encoder 建模时空关系
- Decoder先预测未来手轨迹，再预测未来contact points，并用C-VAE来表达未来不确定性

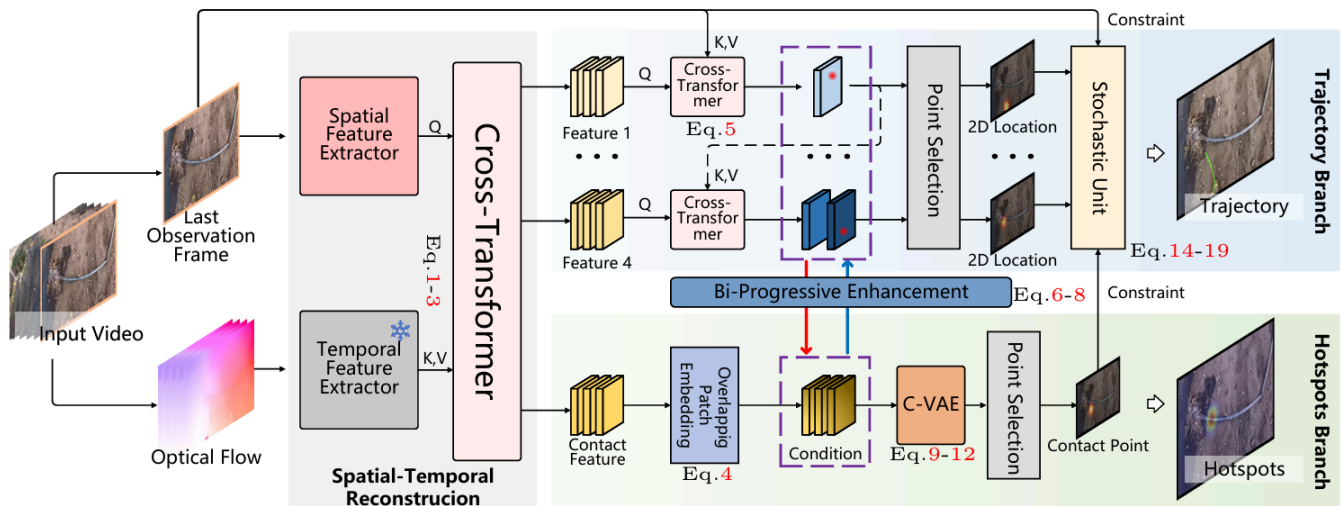


Fig. 3: Overview of the Bidirectional Progressive Transformer. It follows a dual-branch structure with a Bi-Progressive Enhancement Module between them to anticipate future hand trajectories and interaction hotspots.

这篇论文提出trajectory 和 hotspot 应该双向互相修正，手的运动方向可以提示未来接触区域；反过来，未来接触区域也可以约束手应该怎么运动。

- 用 Spatial-Temporal Reconstruction Module 融合：最后一帧的空间信息+整段视频的时间/运动信息
- 分成两个分支：trajectory branch和hotspot branch，用trajectory feature 修正 hotspot feature，hotspot feature 再修正下一步 trajectory feature，然后继续迭代

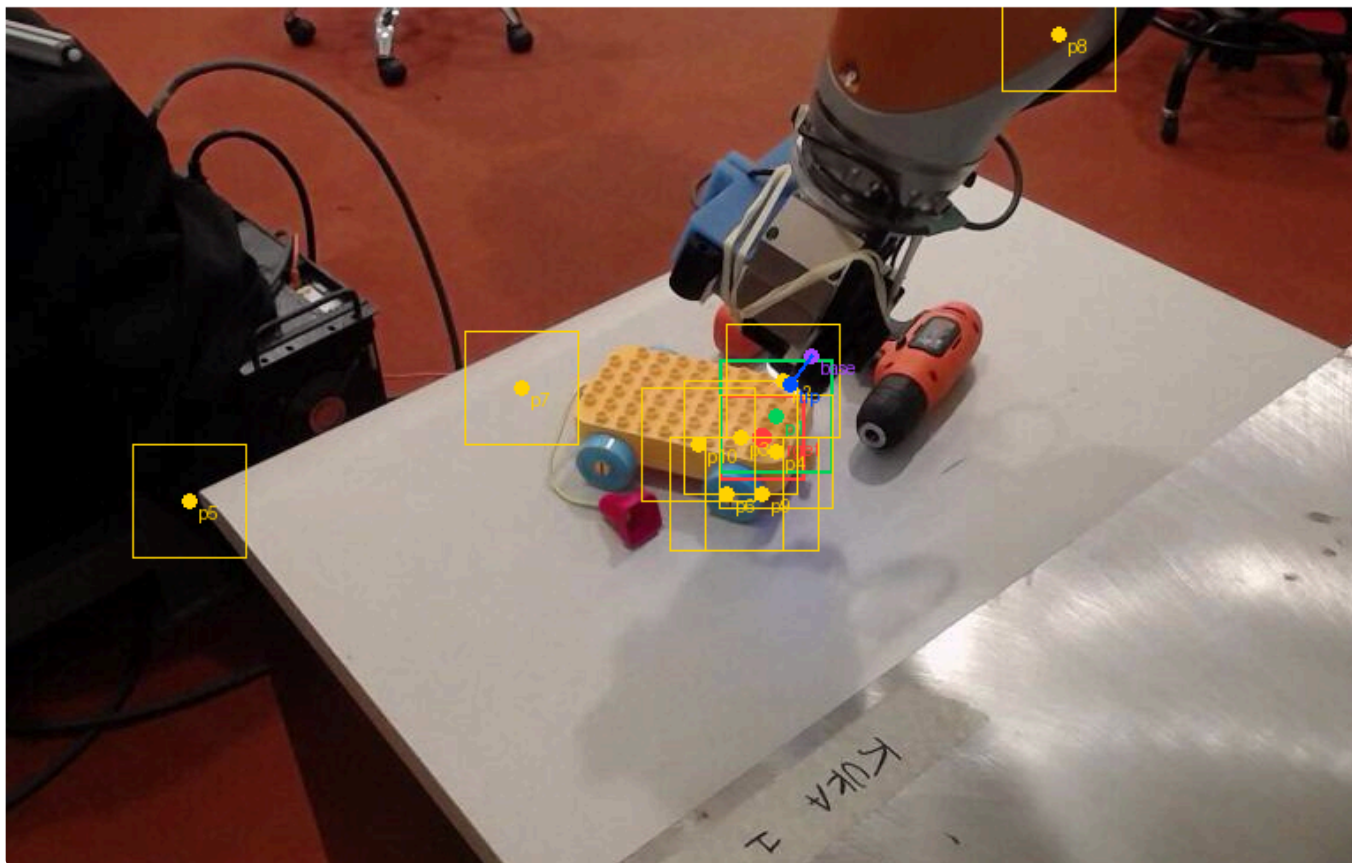
前段时间的工作

- 先做了sensor localizer，手动标了少量sensor_tip, sensor_base，训练了一个轻量 localizer，这一步给模型提供了传感器在哪里，朝向。
- 训练数据
输入：连续的RGB图像，每帧的sensor_tip_xy,sensor_base_xy,sensor_direction_xy
输出：contact frame上的sensor_tip_xy
训练目标：future contact heatmap 训练时，把真实target point变成一张Gaussian heatmap，模型输出也是一张heatmap

RGB sequence + sensor geometry
-> future contact heatmap on current frame

最终从heatmap中取top-K proposal

0_rec_00000_frame000211_seq4 ttc=85 top5_hit=True top5_s64=True



- 主要指标:

median error: 4.0 px

PCK@48: 96.8%

bbox hit: 95.5%

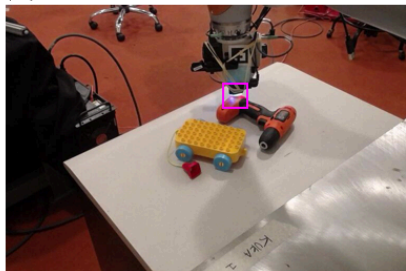
top5 bbox hit: 99.4%

如果只看“未来接触区域是否在proposal中”，效果已经还不错了

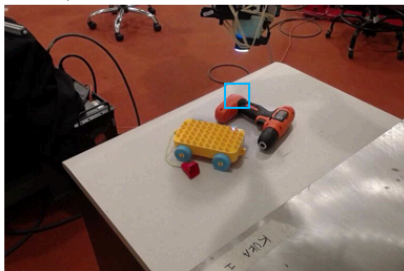
给定一个val样本中的proposal，系统从train cache里找一个最相似的历史接触样本，并把两边的视觉crop和触觉图拿出来对比

learned_nnlis oct_top1 q=0_rec_00011_frame000254_seq4 ret=0_rec_00013_frame000235_seq4 MAE=0.0024 SSIM=0.9984

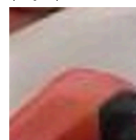
query current frame=254 ttc=5



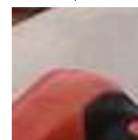
retrieved crop source frame=135 ttc=105



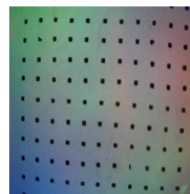
query crop



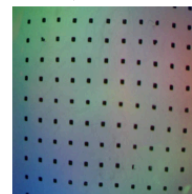
retrieved crop



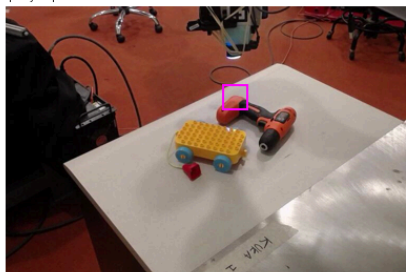
query haptic



retrieved haptic

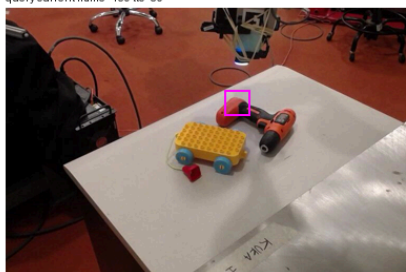


query crop source frame=154 ttc=105

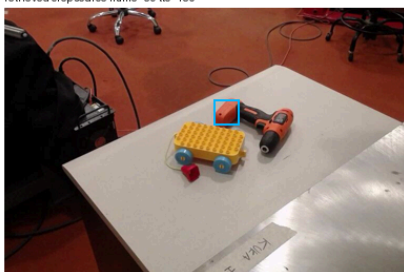


learned_nnlis oct_top1 q=0_rec_00015_frame000159_seq4 ret=0_rec_00023_frame000139_seq4 MAE=0.0069 SSIM=0.9551

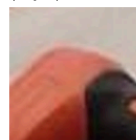
query current frame=159 ttc=80



retrieved crop source frame=39 ttc=180



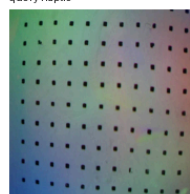
query crop



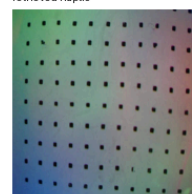
retrieved crop



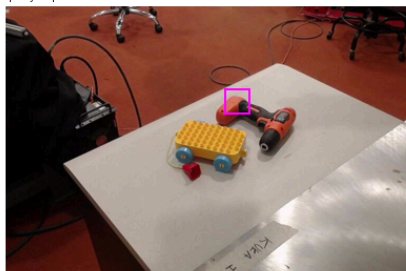
query haptic



retrieved haptic

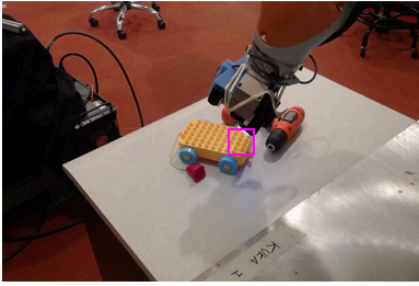


query crop source frame=59 ttc=180

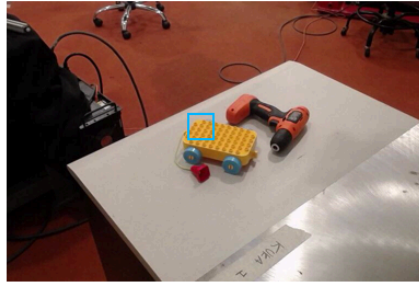


learned_nrlsct_top1 q=0_rec_00000_frame000211_seq4 ret=0_rec_00003_frame000144_seq4 MAE=0.0101 SSIM=0.9565

query current frame=211 ttc=85



retrieved crop source frame=44 ttc=185



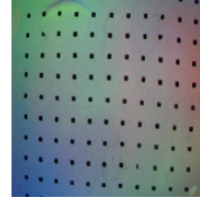
query crop



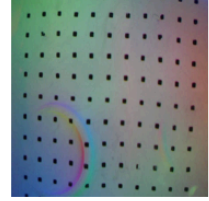
retrieved crop



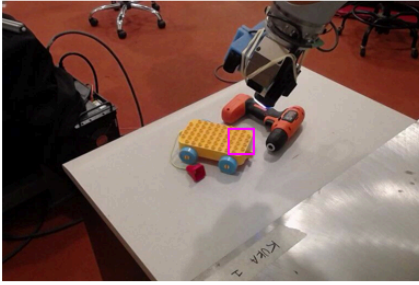
query haptic



retrieved haptic



query crop source frame=111 ttc=185



当前问题：确实能找到同一个物体，但是可能是不同的区域，视觉上不完全错，还做不到精确地区分同一物体上的具体接触位置

但现在只做了最小闭环实验，输入只给了当前帧，和向前3帧，用的U-Net训练，后续考虑更长的时间窗口，并加上trajectory和hotspot的相互约束条件，会换成Transformer